

Review of downstream use of *Masked Autoencoders Are Scalable Vision Learners*

Citation landscape

The target paper, *Masked Autoencoders Are Scalable Vision Learners*, introduced a simple but influential visual self-supervised learning recipe: randomly masking a high fraction of image patches, encoding only the visible patches with a ViT encoder, and using a lightweight decoder to reconstruct missing pixels [1]. In the reviewed citation set, 195 citing records were classified, of which 42 were labeled as confirmed-use candidates. The evidence indicates that the paper’s influence is broad but heterogeneous. Some citing works directly implement MAE-style pretraining or benchmark MAE initializations, whereas others use the paper more conceptually—as support for masked reconstruction, self-supervised pretraining, or scalable representation learning. For that reason, this review distinguishes explicit methodological uses from papers that should be checked manually before being counted as direct uses.

Confirmed uses, extensions, and applications

The most direct line of uptake is in video and spatiotemporal representation learning. BEVT extends masked image modeling into video transformers by separating spatial representation learning from temporal dynamics learning, first applying masked image modeling to images and then combining masked image and masked video modeling on video data [2]. Related works adapt the MAE family to richer audiovisual or video settings: AV-MaskEnhancer builds on ImageMAE and VideoMAE to improve video representations by adding audio-visual masking [11], a state-space VideoMAE variant replaces or augments the VideoMAE framework with bidirectional state-space modeling for efficient action classification [12], and Echo-Vision-FM uses VideoMAE-style pretraining for echocardiogram video foundation modeling [20]. In behavioral science, hBehaveMAE adapts masked autoencoding to motion-capture/behavioral sequence representation, indicating that the MAE paradigm has been transferred beyond static vision into hierarchical behavioral data [22].

A second cluster extends the target paper’s masked reconstruction logic to nonstandard spatial domains. Point-BERT explicitly states that its masked point modeling framework for 3D point clouds was inspired by MAE’s masking-and-reconstruction strategy for image patches [18]. Other works apply MAE-like reconstruction to 3D or volumetric scientific data, including a masked-autoencoder-based three-dimensional foundation model for vortex identification [8] and a transformer-based masked autoencoder for reconstructing pollutant and meteorological fields from sparse mobile-sensing observations [30]. These papers suggest that MAE’s core idea—learning representations by reconstructing structured missing observations—has been generalized from 2D images to 3D geometry, physical simulation fields, and environmental sensing.

A third major use case is domain-specific image pretraining, especially where labeled data are scarce. Remote-sensing and Earth-observation studies use masked image modeling to pretrain ViTs for remote-sensing scene classification and SatViT-style Earth-observation foundation models [4,21]. Agricultural classification work similarly pretrains ViTs on large agricultural image collections using masked image modeling before downstream classification [14]. Histopathology and biomedical imaging have also adopted the MAE/MIM family: one histopathology scaling study frames masked image modeling as a central self-supervised strategy for pathology foundation models [16], a cervical cytology screening study reports MAE-

based self-supervised pretraining on unlabeled cervical images [25], BrainMRIFM evaluates MAE-ViT among slice foundation models for brain MRI [27], and SLOTMFound uses a retinal foundation-model pipeline grounded in the MAE paradigm [29]. These applications are consistent with the target paper's original claim that masked reconstruction can provide transferable visual representations [1].

Several papers use MAE as a benchmark, initialization, or comparison point rather than as the main proposed method. A detection-transfer benchmark evaluates MAE as one of several ViT initializations for Mask R-CNN transfer to COCO [17]. PeCo compares against MAE in masked image modeling and reports improved transfer performance using a perceptual codebook target [19]. A healthcare foundation-model evaluation includes MAE-trained ViTs among a large set of general-purpose vision embeddings used for downstream medical prediction tasks [24]. These benchmarking papers are important because they show that MAE became not only a method to extend, but also a reference point for assessing new vision foundation models.

The evidence also shows diffusion into scientific, biomedical, and multimodal signal domains. A robotics representation-learning paper uses masked autoencoding as part of its analysis of visual representations for visuomotor and language-conditioned robot-learning tasks, contrasting masked autoencoding's low-level spatial strengths with contrastive methods' semantic strengths [3]. scMMAE adapts masked autoencoding with cross-attention for single-cell multimodal omics fusion [7]. A foundation model for fNIRS classification uses a self-supervised masked autoencoder framework for few-shot neural-signal classification [9]. RamanMAE applies masked autoencoding to Raman spectral data for biologically meaningful molecular-imaging representations [28]. Other domain-specific adaptations include a quaternion-based ViT-MAE for polycrystalline EBSD scans [5], a physics-informed masked autoencoder for optical microscopy [6], masked-autoencoder-based synthetic solar irradiance sequence generation [13], a fading-mask autoencoder for distributed fiber acoustic sensing [26], and an MAE-based image inpainting-steganography method that explicitly builds on the original MAE architecture [31]. Collectively, these studies indicate that the target paper's influence is not limited to natural-image classification; its masking, asymmetric reconstruction, and scalable pretraining principles have been repurposed across materials science, microscopy, omics, physiology, spectroscopy, environmental sensing, and information hiding.

Limitations of this draft

This synthesis is based on the evidence supplied in the prompt: metadata, abstracts, classifier audit fields, and extracted full-text snippets. The evidence base is therefore mixed. Some papers have explicit full-text evidence showing direct use of MAE, such as Point-BERT, the detection-transfer benchmark, PeCo, Echo-Vision-FM, MC-SSL0.0, the cervical cytology study, SLOTMFound, the air-pollution reconstruction study, and MAE-InpaintSteg [17–20,23,25,29–31]. Other records are supported only by abstracts, so their precise relationship to the target paper should be treated as provisional until the full text is checked [2–9,11–15,21,22,26,28]. Finally, several records labeled as “yes” contain contradictory or indirect evidence; these are not treated as confirmed methodological uses below and should be manually reviewed.

Papers that should be manually checked next

- **Domain-PFP allows protein function prediction using function-aware domain embedding representations** — the classifier reason itself states that the full text shows no evidence of using the target MAE paper; this should likely be reclassified rather than counted as a confirmed use. DOI: <https://doi.org/10.1038/s42003-023-05476-9>

- **PillarNeSt: Embracing Backbone Scaling and Pretraining for Pillar-Based 3D Object Detection** — the evidence supports broad use of image-domain backbone pretraining/scaling, but does not clearly establish use of MAE specifically. DOI: <https://doi.org/10.1109/tiv.2024.3386576>
- **A robust multi-scale clustering framework for single-cell RNA-seq data analysis** — the evidence suggests comparison to or citation through scMAE/masking strategy, but direct use of the target paper is unclear. DOI: <https://doi.org/10.1038/s41598-025-03603-6>
- **Advancing animal behavior recognition with self-supervised pre-training on unlabeled data** — the full-text evidence points to TF-C and PatchTST rather than MAE-style masked reconstruction. DOI: <https://doi.org/10.1038/s41598-025-27736-w>
- **VIRL: Volume-Informed Representation Learning towards few-shot manufacturability estimation** — the extracted evidence appears to cite MAE as background for self-supervised inpainting rather than directly applying MAE. DOI: <https://doi.org/10.1007/s10845-025-02575-8>
- **Dual stage MRI image restoration based on blind spot denoising and hybrid attention** — the evidence distinguishes MAE-style image completion from blind-spot denoising but does not clearly show that the proposed method uses MAE. DOI: <https://doi.org/10.1186/s12880-024-01437-8>
- **Pretraining the Vision Transformer Using Self-Supervised Methods for Vision Based Deep Reinforcement Learning** — the abstract mentions several state-of-the-art self-supervised methods but does not explicitly name MAE. DOI: <https://doi.org/10.3233/faia230359>
- **Natural Scene Text Editing Based on AI** — the evidence appears to describe MAE as background rather than applying it in the proposed text-editing model. DOI: <https://doi.org/10.48550/arxiv.2111.15475>
- **Improving generalizability and data efficiency for MHC-I binding peptide predictions through structure-based geometric deep learning** — the evidence references MAEs as examples of self-supervised learning, but direct adaptation of the target method remains uncertain. DOI: <https://doi.org/10.21203/rs.3.rs-3924124/v1>
- **Comment on egusphere-2023-2261** — the evidence is conceptual background about corrupted-data reconstruction and masked autoencoders, not a direct use of MAE. DOI: <https://doi.org/10.5194/egusphere-2023-2261-rc2>
- **Detection of Malaria Infection from parasite-free blood smears** — the snippet mentions MAE among self-supervised models, but the available evidence does not clearly show that MAE was one of the evaluated backbones. DOI: <https://doi.org/10.64898/2025.12.29.25343125>
- **PhysioBridge: Physiology-Constrained Self-Supervised Foundation Model for Cross-Device ECG-PPG Learning with Conformal Risk Control** — the abstract mentions coupled masked modeling, but not MAE specifically. DOI: <https://doi.org/10.1145/3797491.3797535>

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